

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-19 02:40 UTC

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Generated 2026-05-19 02:40 UTC

- Cycles recorded: **813**
- Time span: 605.0h (~1.34 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	773
BARRIER_HIT	20
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-19 00:11 UTC	INCONCLUSIVE	Decision Tree Depth and Communication Complexity via Lifting
2026-05-19 00:24 UTC	INCONCLUSIVE	Laplacian Apolar Rank Caps Delorme-Poljak Max-CUT
2026-05-19 00:32 UTC	INCONCLUSIVE	SoS-2 Gap Cross-Read Kronecker Sum
2026-05-19 00:39 UTC	INCONCLUSIVE	Rank Lower-Bounds Read-Twice BP for IP ₂ SoS Cone-Gram Stable Rank
2026-05-19 00:46 UTC	INCONCLUSIVE	Caps AC ⁰ Depth-d Size for PARITY
2026-05-19 01:09 UTC	INCONCLUSIVE	Talagrand L1 Influence Spread of Clause-Falsification Polynomial
2026-05-19 01:23 UTC	INCONCLUSIVE	Border Rank Lower Bound for Communication Complexity of Disjointn
2026-05-19 01:33 UTC	INCONCLUSIVE	Communication Complexity Lower Bound for Disjointness Implies Mon
2026-05-19 01:40 UTC	INCONCLUSIVE	Cubical First-Betti Linearly Lower-Bounds DNF-MCSP
2026-05-19 01:51 UTC	INCONCLUSIVE	Fourier Spectral Entropy of Clause-Falsification Polynomial Bound
2026-05-19 02:06 UTC	INCONCLUSIVE	2-Adic Cokernel Type of Sign Matrix Caps Randomized CC
2026-05-19 02:13 UTC	INCONCLUSIVE	vdW 3-AP Density of MAJ-Lifted NW Designs Tracks Overlap k Lorentzian Defect of Cut Polynomial Lower-Bounds Max-CUT SoS-2 Ga

Time	Verdict	Title
2026-05-19 02:22 UTC	INCONCLUSIVE	Level-1 Fourier Mass Fraction of Term Family Bounds
2026-05-19 02:31 UTC	INCONCLUSIVE	Monotone CLIQ
2026-05-19 02:40 UTC	INCONCLUSIVE	Dilworth Width of Gate-Cone Poset Lower-Bounds AC^0
		PARITY Size
		Mean Log-Bias of Internal Gates Lower-Bounds AC^0
		PARITY Size

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```